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Crystal Gardens

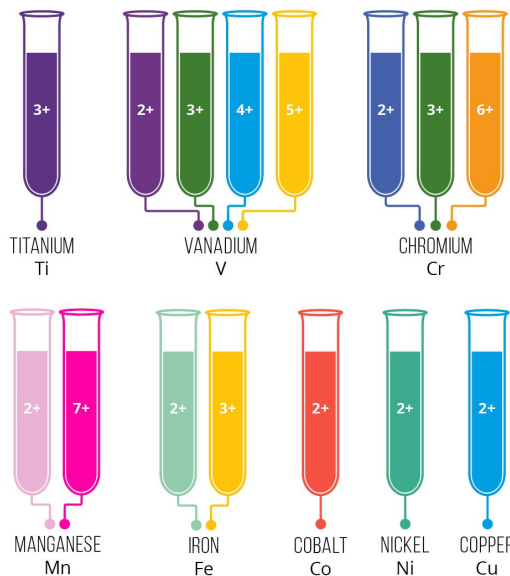
An Overview



How Does it Work?

Water glass $\rightarrow$ sodium silicate Na_2SiO_3

The metals salts we use are **transition metals**.



Periodic table showing elements and their atomic numbers. An arrow points to the transition metal block (d-block).

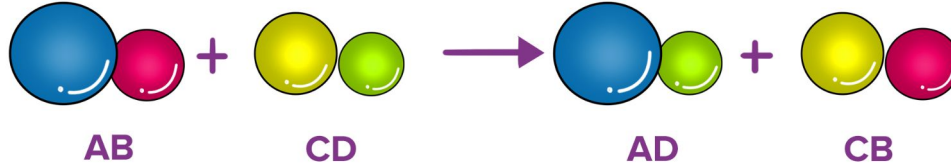
We like transition metals because they are **colourful**.



Lanthanide and actinide series of elements.

How Does it Work?

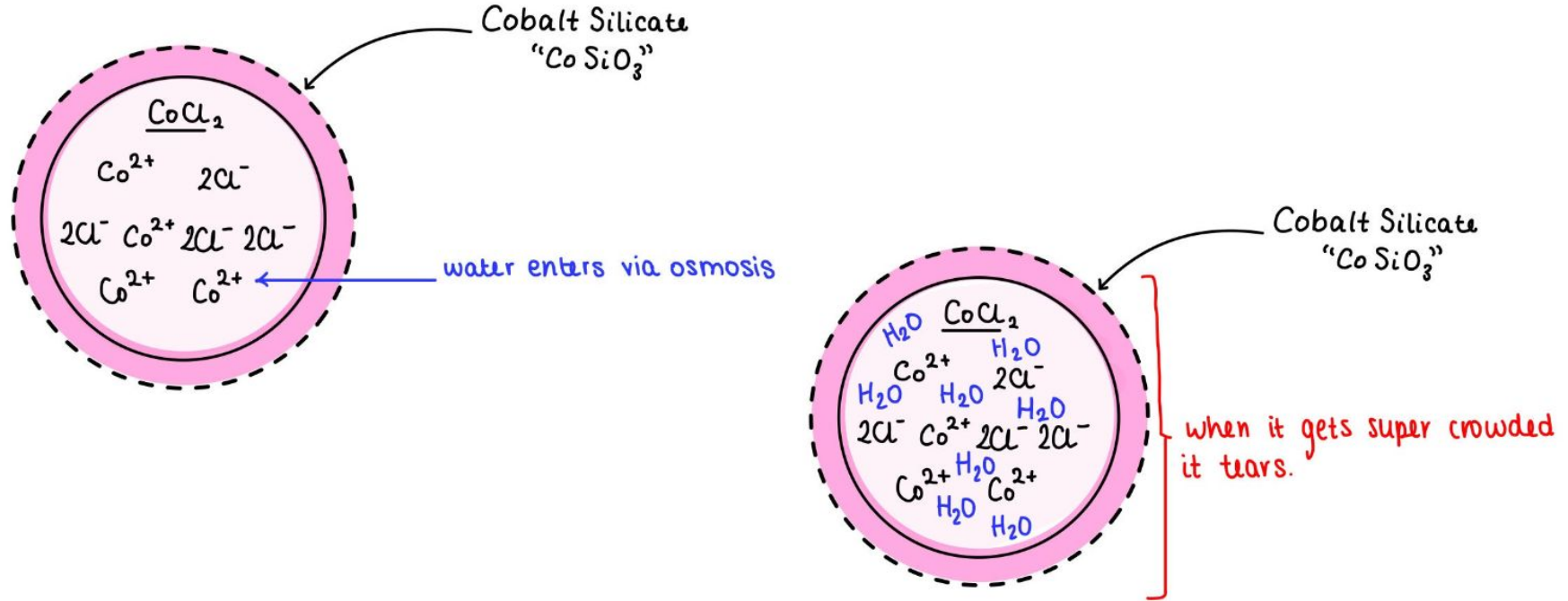
- When we add the **transition metal salts**.
 - i.e. cobalt chloride, CoCl_2
- It dissolves.
- A **displacement reaction** occurs and a transition metal silicate is formed.



- i.e. cobalt silicate CoSiO_3
- Most transition metal **silicates** are **insoluble** → **do not** dissolve.

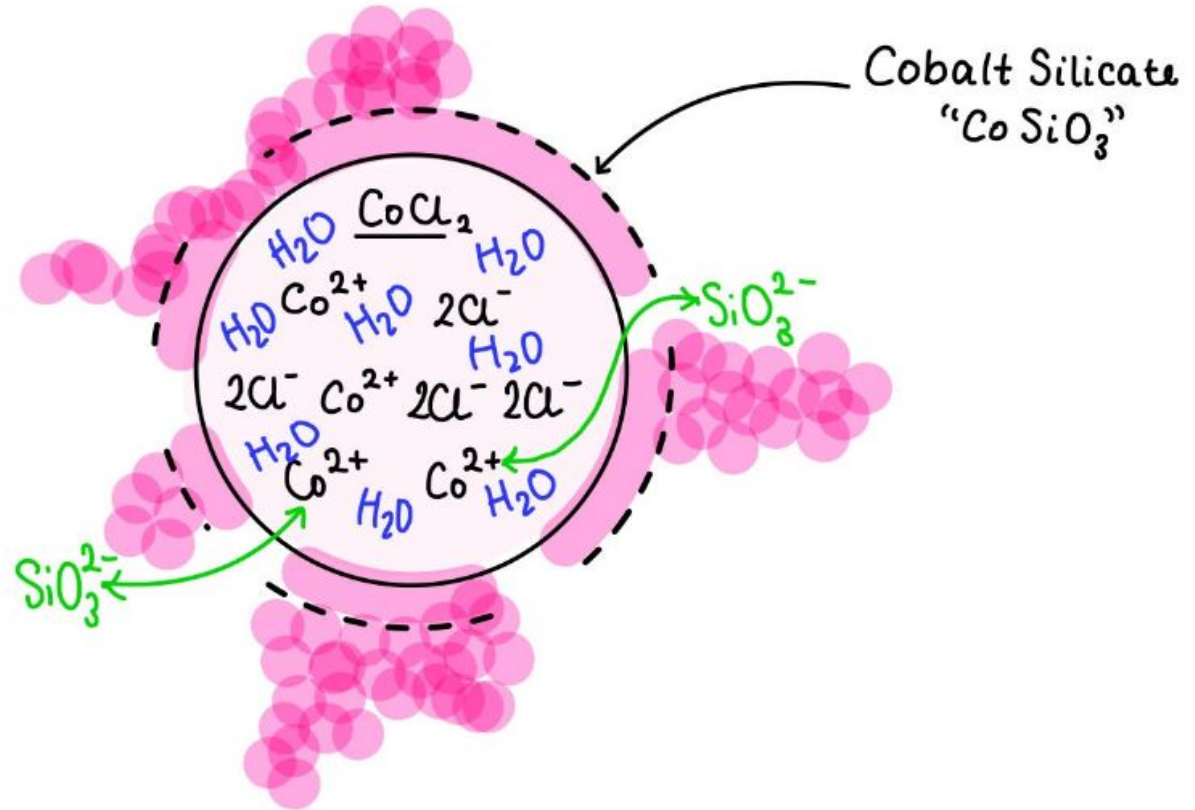
How Does it Work?

- Cobalt silicate is a **semipermeable membrane**.
 - Acts as a **barrier** that allows **certain but not all** substances to pass.



How Does it Work?

- This cycle repeats itself and crystals form on the outside continuously until all the CoCl_2 has reacted.



Method:

1. $\frac{1}{4}$ fill a **glass jar** with **water glass** solution and top it up with water.
2. Mix it up well with a **glass rod**.
3. Using the **forceps**, take a large **crystal** of each of the solids provided and carefully drop it into the water glass.
4. Watch carefully.
5. Put the **lid** on the jar and leave for a week for full development.

Try not to disturb the glass jar while the crystals are growing.

